



Improving medical data quality via synthetic data generation: a review

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Abstract

Most Artificial intelligence (AI) algorithms are trained on large-scale datasets to learn complex, underlying patterns to make informed decisions. However, due to their data-intensive nature, the performance of these algorithms highly depends on the quality of the training data. Due to strict privacy regulations, large medical datasets are not readily available, which leads to reduced data sizes as well as under-representation of some classes and demographic groups. These data shortcomings, if not handled, are replicated by the AI algorithms, thus compromising their performance. One potential solution to this problem is the augmentation of data by generating synthetic samples that possess the same real-world data properties. Therefore, this study explores the synthetic data generation process and pre-existing research, mainly focusing on statistical, probabilistic, and Machine Learning (ML) based tabular data generation methods. For this purpose, we analysed high-quality peer-reviewed articles extracted from different databases. The findings show that synthetic data not only increases data volume but also improves its quality by increasing diversity and enhancing the representation of various demographic groups within the datasets. Synthetic Minority Oversampling Technique (SMOTE) and its variants are the commonly used techniques with up to 12% improvement reported in the classification performance of a classifier. However, the quality of artificial data depends on the underlying data generation process, the characteristics of real-world data used for subsequent modelling, and the evaluation metrics to assess data quality.

Keywords Synthetic data generation · Artificial data · Statistical and probabilistic · Healthcare · Medical datasets · ML

1 Introduction

In healthcare, the integration of big data with AI techniques can be used for disease diagnosis, developing personalised healthcare systems, and aiding clinicians in decision-making (Price and Nicholson 2019; Alowais et al. 2023). The AI-enabled healthcare systems can significantly enhance the provision of healthcare and other services that promote human health and well-being. For example, an AI diagnosis system created by medical professionals at London's Moorfields Eye Hospital can prescribe therapies for over 50 eye conditions with a 94% accuracy (Lee and Yoon 2021). The Mayo Clinic in the United States (US) has employed AI to screen cervix cancer to detect pre-cancerous changes. Researchers state that the algorithm trained on 60,000 cervical images is 22% more accurate than a human expert (Arsene 2024).

A large number of these AI algorithms are data-driven in nature; they are trained on large datasets to learn hidden patterns based on which they make decisions (Alowais

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et al. 2023). Therefore, the performance of these algorithms depends on the quality of the data being fed into them (Ntoutsis et al. 2020). There is no general agreement on the dimensions of high-quality data. But the most commonly found in the literature are described below:

- Accuracy can be defined as the degree to which data is correctly represented (Sidi et al. 2012). Data is considered accurate if it is free from errors and uncertainties and corresponds to the real-world and real state of information (Ridzuan and Zainon 2024).
- Consistency refers to the representation of data in a uniform structure, format, and labels (Sidi et al. 2012). It is a measure of data equivalency in formats, values, and domain across different data sources and throughout data processing stages (Cai and Zhu 2015).
- Completeness is the ability to represent every state, group and class of real-world objects (Batini et al. 2009). A dataset is considered complete if it covers sufficient information in depth, breadth and scope (Liu et al. 2017).

The integration of high-quality data with AI can be used to build systems that can assist in disease diagnosis, develop personalised healthcare systems, and aid clinicians in decision-making (Alowais et al. 2023; Price and Nicholson 2019). However, the complexity and sensitivity of medical data, coupled with strict privacy regulations, can result in data scarcity, restricting AI from reaching its full potential (Sufi 2024). Medical data scarcity leads to several constraints, including non-uniform, unlabelled, inadequate or poor-quality datasets (Gangwal et al. 2024). These training data can result in unreliable AI models that ultimately make poor decisions (Budach et al. 2022). Synthetic data generation is a potential solution to improve the quality and combat the scarcity of healthcare data (Lu et al. 2023). Synthetic data can be defined as data that has been artificially generated from real-world instances. It exhibits the same statistical properties as real-world data (El Emam et al. 2020). In the healthcare domain, data may contain sensitive information that needs to be dealt with in such a way as to preserve privacy without compromising the quality of the data. As synthetic data is generated rather than collected, the problem of revealing sensitive information can be minimised (Figueira and Vaz 2022). Also, synthetic data, if carefully generated, can be of high quality, which can increase the AI model's performance. Furthermore, it can also help to eliminate biases to develop more generalised solutions (Kortylewski et al. 2019).

Medical data is available in a variety of formats, i.e., images, text or tabular data, audio, and sensor data. There is a lot of advanced research on synthetic data generation in computer vision tasks, i.e., image generation (Figueira and Vaz 2022). As tabular data is the fundamental type

of data modality, it is of immense importance and interest. For this reason, the focus of this study is particularly on techniques of synthetic tabular data generation. Several techniques have been developed to generate artificial data. These techniques are categorised into two domains: standard techniques and Deep Learning (DL) techniques, as shown in Fig. 1. Deep Learning (DL) models include Generative Adversarial Network (GAN) (Shahul Hameed et al. 2024), Autoencoders (Pathare et al. 2023), Long-Short Term Memory (LSTM) (Abd-Alhussain et al. 2024), Recurrent Neural Network (RNN) (Sutskever et al. 2011), and many more. The most widely used DL technique is GAN, with several variants being developed, such as Tabular GAN (TGAN), Conditional Tabular GAN (CTGAN), Medical GAN (MEDGAN), and Multi-Categorical MEDGAN (MC-MEDGAN), etc (Hameed et al. 2024). GAN is a type of neural network architecture where two networks, a generator and a discriminator, are trained simultaneously (Anaissi et al. 2023). The generator is responsible for producing fake data, and the discriminator evaluates and tests the validity of the generated data. DL models are referred to as black box; therefore, for critical applications, including medical device software, these models impose challenges for explainability and transparency as to how the data is being generated. Moreover, DL models take raw data as input and automatically learn features; due to which these models provide minimal control over the parameters and the data generation process (Fazi 2021; Von Eschenbach 2021). These architectures, therefore, can hinder the trustworthiness of these models, which is a prerequisite for AI implemented in critical sectors, including the healthcare industry (Von Eschenbach 2021). Additionally, DL models require high computational resources that has an adverse effect on the environment due to excessive heat generation (Ahmed et al. 2023). Also, several studies have reviewed DL models for synthetic data generation (Pezoulas et al. 2024; Figueira and Vaz 2022). Consequently, our study focuses only on standard techniques for generating synthetic tabular data. This literature review discusses various statistical, probabilistic and ML techniques, providing a comparative review and investigating the opportunities and limitations of these techniques for medical data generation.

1.1 Motivation

Large tabular healthcare datasets are being collected electronically by hospitals, and public and private healthcare providers. Despite being a valuable resource for scientists, this medical data is not easily available to the research community due to its highly sensitive nature and privacy issues, thus leading to data scarcity (Goncalves et al. 2020). AI models are data-driven in nature; therefore, their output

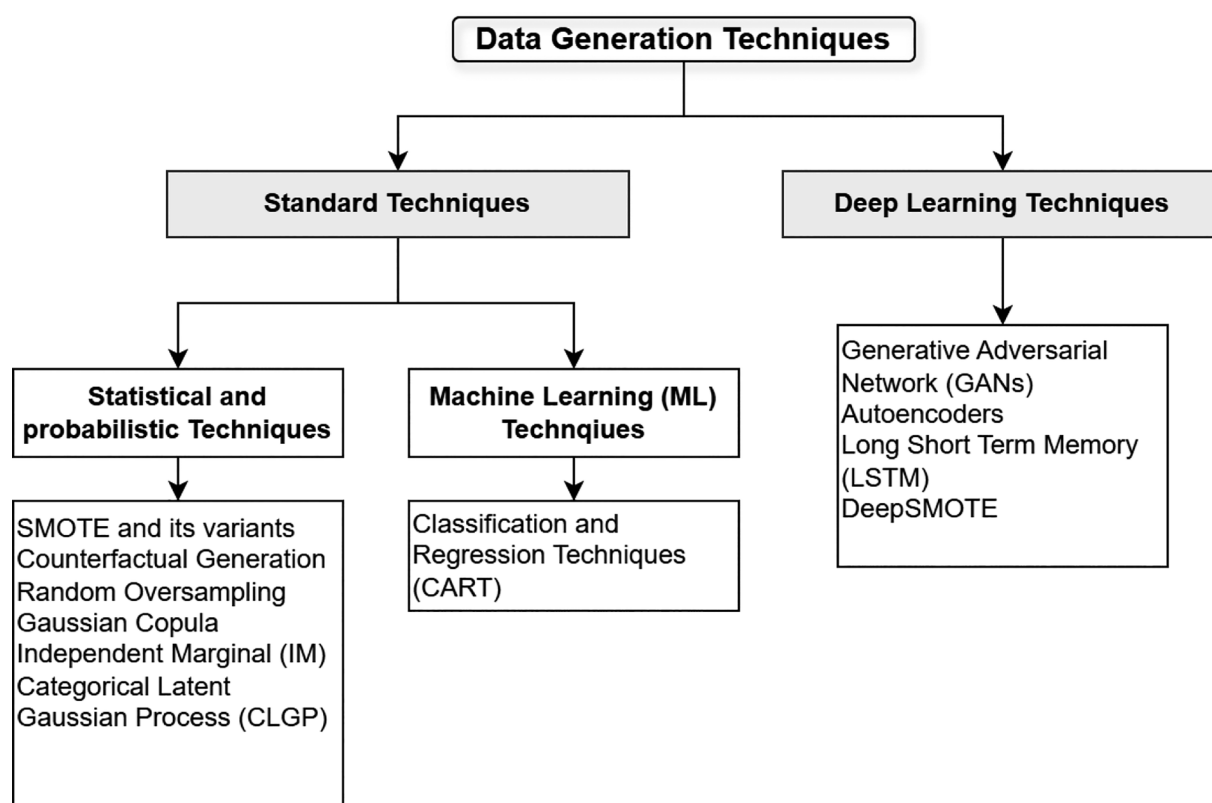


Fig. 1 Classification of data generation techniques

highly depends on the quality of the data being fed into them. In healthcare, AI systems can be used to assist clinicians in diagnosis or treatment planning, hence any uncertainties can risk human life. For example, suppose an AI model for breast cancer detection is trained on a dataset that lacks the representation of African-American women who have different breast densities. In that case, the algorithm will produce ill-suited recommendations for the underrepresented group (Babic et al. 2019). Low-quality medical data may reduce the effectiveness of AI models in healthcare applications, which can prevent the wider adoption of data-driven technologies that could otherwise have the power to transform the healthcare industry (Sufi 2024; Ibrahim et al. 2021). To address this problem, researchers are now focusing on generating synthetic data that may exhibit the characteristics of real-world data (Ali et al. 2023). Data generation techniques help combat data scarcity while improving data quality and AI model performance, preserving privacy, and help deflate inherent biases to provide more generalised solutions (Abay et al. 2019; Jaipuria et al. 2020). Therefore, this paper explores the literature to analyse different synthetic data generation techniques. The devised research questions are as follows:

RQ1. How is synthetic data generated?

RQ2. What are the most common standard techniques for tabular synthetic data generation?

RQ3. What is the difference in the performance of different techniques?

The rest of the paper is divided into six subsections. Section 2 explains the search methodology to conduct this study. Section 3.1 provides an overview of the synthetic data generation process. Furthermore, Sect. 3.2 discusses the most commonly used data generation techniques in detail. Section 3.3 provides a comparative analysis of different techniques on commonly used medical datasets. Section 4 explains the strengths, weaknesses and implications of our study. Lastly, Sect. 5 concludes the overall discussion.

2 Search methodology

To conduct this review, several databases were explored, including IEEE Xplore, Google Scholar, Science Direct, PubMed and ACM Digital Library, to extract research articles relevant to tabular synthetic data generation. These databases were chosen for their large collection of peer-reviewed technology and application-based research studies. The following search query was formulated to search these different databases:

((Synthetic Data Generation OR Artificial Data OR Oversampling OR Imbalance data) AND (Artificial Intelligence OR Machine Learning OR statistical OR Probability) AND (text OR tabular dataset OR medical data))

Slight adjustments were made to the query to improve the retrieved results. Also, the following exclusion and inclusion criteria were devised to select the most relevant studies.

2.1 Inclusion criteria

- Research studies focused on tabular synthetic data generation techniques were included.
- Articles published between 2020 and March 2025 were considered to review recent advances in the field of medical data generation because review articles including (Azhar et al. 2022) and (Figueira and Vaz 2022) have already reviewed synthetic data generation techniques in previous years.
- Only peer-reviewed articles published in conferences, journals, proceedings, and book chapters were included.
- Only research studies written in English were included.

2.2 Exclusion criteria

- Studies that do not discuss the generation of synthetic medical datasets were not selected.
- Articles that do not align with our devised research questions were not selected, i.e. studies that are not evaluated on at least one medical dataset were not included.

Figure 2 highlights the methodology used to conduct this literature review.

The identification stage includes the collection of a total of 562 records from different databases, i.e., Google Scholar, PubMed, IEEE, Science Direct, and ACM. Preliminary screening by applying filters for year selection (2020–2025) was applied, choosing studies that are peer-reviewed and written only in English, resulting in 116 articles. Subsequently, the distinct records underwent a preliminary screening process based on their titles and abstracts to eliminate irrelevant research based on exclusion and inclusion criteria, resulting in 45 studies. During the eligibility phase, full-text articles of the screened records are carefully examined following predetermined standards to make sure that only the studies that meet the review's scope and quality requirements are included. Finally, we selected 27 articles for detailed analysis and review. Figure 3 shows the year-wise distribution of our selected research articles.

3 Results

This section discusses our findings in order of the devised research questions as stated in Sect 1.1.

3.1 RQ1. How is synthetic data generated?

The healthcare industry is collecting and producing large amounts of data, including tabular, text, images, audio, and sensor data that open many opportunities but also lead to challenges at the same time (Dash et al. 2019). In high-stakes sectors like healthcare, under-represented datasets can have adverse effects on susceptible communities and situations (Sambasivan et al. 2021).

Despite being a valuable resource for scientists, limited medical datasets are publicly available to the large research community due to their highly sensitive nature and privacy issues (Goncalves et al. 2020). Therefore, the unavailability of high-quality, well-represented healthcare data may hinder the broader use of data-driven technologies, which could otherwise potentially reform the healthcare sector (Ibrahim et al. 2021). To address this, a large proportion of research is now focusing on generating synthetic data that may exhibit the characteristics of real-world data (Ali et al. 2023).

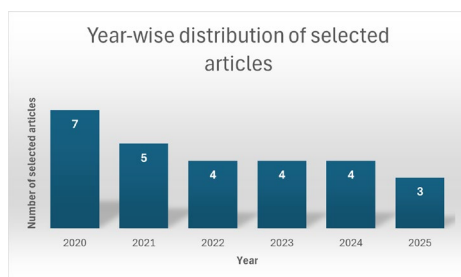
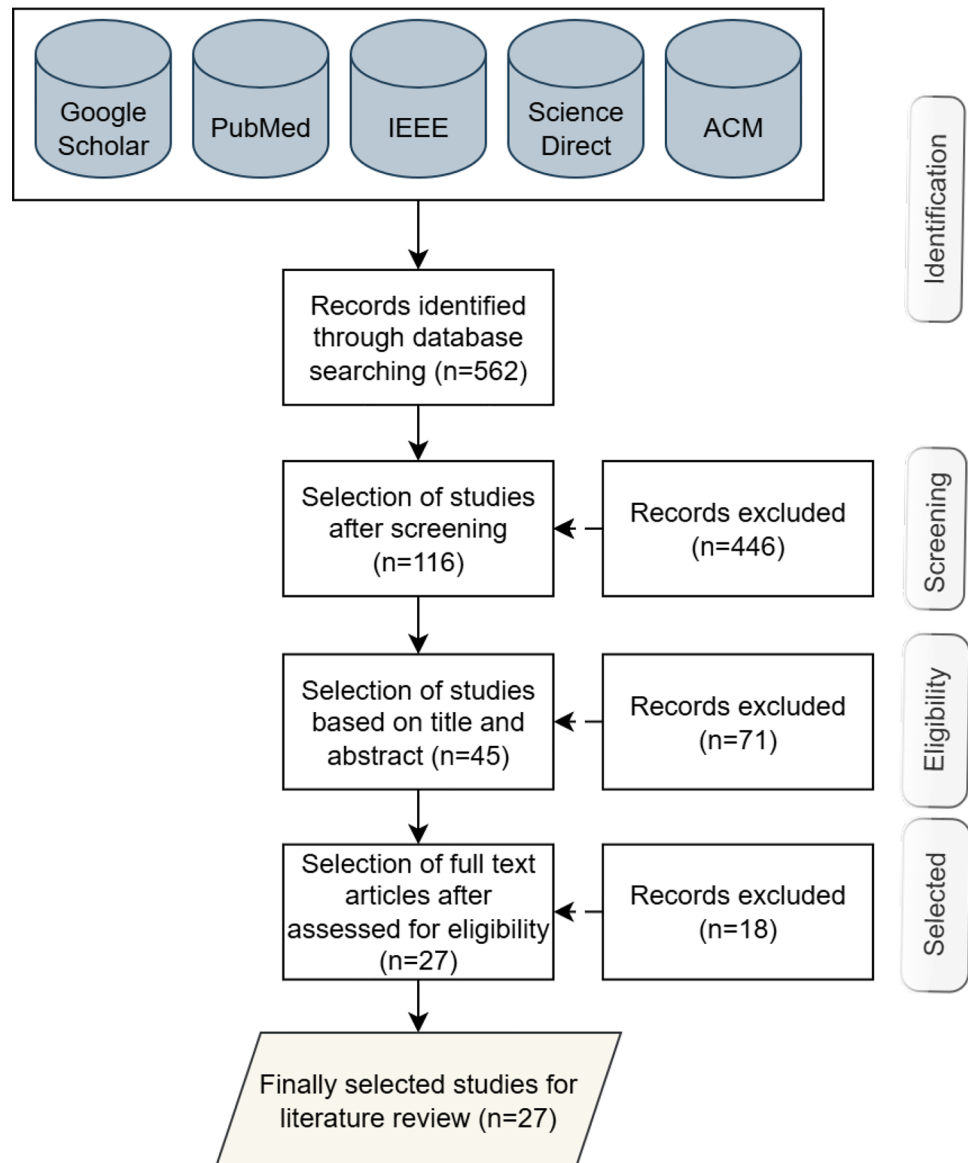
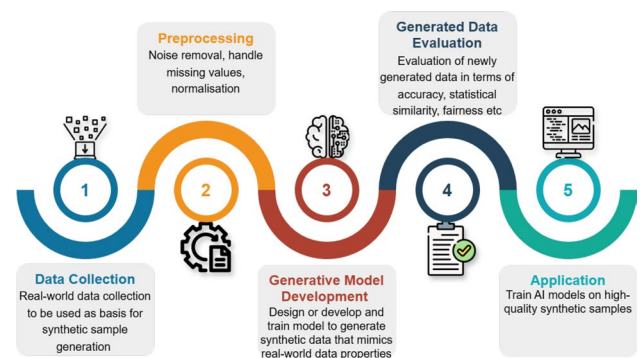
Synthetic data can be defined as data that has been artificially generated from real-world data and exhibits similar statistical properties (Shahul Hameed et al. 2024). Synthetic data can help mitigate several challenges, including:

- As synthetic data is generated rather than collected, the problem of revealing sensitive information or identity disclosure can be minimised (Figueira and Vaz 2022).
- Synthetic data not only helps to secure patient anonymity but also increases relevant data volume and accessibility, thus mitigating high data cost and data scarcity issues (Pezoulas et al. 2024).
- Synthetic data, if carefully generated, can be accurate, consistent across labels, formats and values that can increase the AI model performance.
- It can also help to eliminate the risk of biases by handling imbalances in the datasets to enhance completeness, and to help develop more generalised and fair solutions (Kortylewski et al. 2019).

Figure 4 shows the general workflow of the synthetic data generation process. Each of these steps is defined below:

3.1.1 Data collection

The first step is to collect real-world datasets that form a basis for synthetic data generation. The quality of generated artificial data highly depends on the original or real-world datasets used in the generation process (Shahul Hameed et al. 2024).

Fig. 2 Flow diagram of the search methodology**Fig. 3** Year-wise distribution of selected articles**Fig. 4** Workflow of synthetic data generation

3.1.2 Preprocessing

The synthetic data tends to replicate the underlying distribution of the original dataset; therefore, it must be suitable for subsequent modelling. The real-world datasets are pre-processed for missing values, normalisation and transformation to ensure their suitability for further generation (Pezoulas et al. 2024).

3.1.3 Generative model development

The artificial data generation process should be designed not only to increase sample size but to generate meaningful data instances that increase accuracy, consistency and completeness of the dataset by considering the following factors:

Handling class imbalance: Imbalanced datasets are a significant issue in real-world medical datasets. An imbalance occurs when there are many more data samples of one class, called the majority class, compared to the other, called the minority class (Kumar et al. 2022). AI algorithms trained on such datasets fail to capture the underlying pattern in the minority class, leading to poor classification performance (Azhar et al. 2022). Three major data complexity issues accompany imbalanced datasets, including noise, class overlap and class disjuncts (Azhar et al. 2022). Noise can be defined as the presence of a data point in a different class region with a different class label. These points poorly affect the performance of the classifier (Garcia et al. 2015). The other factor that reduces the performance of the classifier is the class overlap. Class overlap or borderline data points can be defined as instances that fall into various classes and are located in an uncertain area because their feature values are similar (Vuttipittayamongkol et al. 2021; Nekooeimehr and Lai-Yuen 2016). Lastly, small disjunct clusters are defined as those that are composed of instances from one class and are situated in the area of another class as a result of under-represented sub-concepts in imbalanced problems (Almutairi and Janicki 2020). Figure 5 shows the illustration of all three data complexity problems.

Synthetic data should be generated by keeping in view the data distribution, complexity and number of data samples for each class in the original dataset.

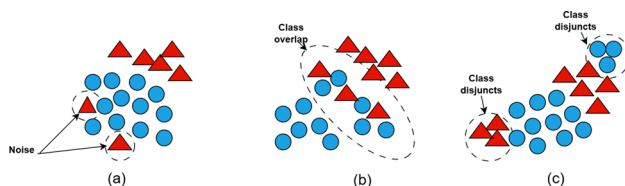


Fig. 5 Data complexities **a** Noise **b** class overlap **c** class disjuncts where blue represents majority class instances, and red represents minority class instances

Handling imbalance in protected attributes: Data oversampling without considering imbalance in the representation of different groups may introduce bias that can be replicated in the outcomes of AI algorithms (Mehrabi et al. 2021). Protected attributes in health data, such as gender, race, and ethnicity, are among the most prevalent sources of bias (Juwara et al. 2024). Disparities across these group attributes often split the population into privileged and underprivileged groups (Yan et al. 2020). The privileged group is more likely to receive favourable outcomes compared to the unprivileged group (Hort and Sarro 2022). AI models trained on datasets containing imbalances in protected attributes can lead to unfair and biased decisions (Jain et al. 2023). For example, in 2016, German scientists used neural networks to create a system for detecting skin cancer. 95% of melanoma instances were correctly detected by the method. It performed better than 58 dermatologists after being trained on 10,000 skin images. Later, it was discovered that the data did not generalise well to a varied population and was heavily dominated by images of light-skinned individuals (Hameed et al. 2024). Therefore, the medical data generation process must consider imbalances in protected attributes to mitigate the risk of potential bias from real-world datasets.

In the literature, various data generation techniques have been proposed. A detail of each method is given in the next section.

3.2 RQ2. What are the most common standard techniques for tabular synthetic data generation?

Table 3 provides an overview of the different data generation methodologies, outlining data complexity handled, employed techniques, and evaluation metrics. Table 4 provides an overview of the medical datasets used by each study to validate its data generation process. The most commonly used datasets include the Pima Indian Diabetes Dataset (Pima), Haberman, Newthyroid1, Parkinsons disease and Wisconsin Breast Cancer Dataset (WDBC) dataset.

This section provides details of different data generation methods employed by selected reviewed articles. Figure 6 presents the proportion of studies discussing Synthetic Minority Oversampling Technique (SMOTE) variants, and other techniques and tools to generate synthetic data. Figure 7 shows an overview of these techniques and their corresponding data complexities addressed.

A large number of studies have proposed the use of SMOTE in conjunction with other techniques. Therefore, we have divided these SMOTE variants based on the focus of each article.

Proportion of studies based on SMOTE and other techniques

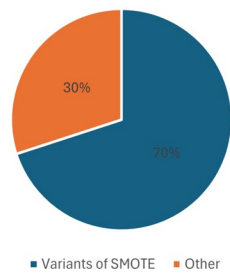


Fig. 6 Proportion of studies using SMOTE and other techniques in each selected article

3.2.1 Synthetic minority oversampling technique

SMOTE is a widely used oversampling method. SMOTE generates synthetic samples by approximating minority class samples. These techniques are designed to handle class imbalance where the data distribution is skewed to one class label and lacks enough information on other class instances. In SMOTE, the distance between a given data point and

its nearest neighbours from the same class is calculated. The distance is multiplied by a random number between 0 and 1 and added to the selected point (Chawla et al. 2002). The traditional SMOTE only generates minority class data points without considering noisy samples, class overlap and small disjuncts. Therefore, SMOTE generates samples in overlapping regions or may use noisy class samples that result in ineffective artificial data samples (Azhar et al. 2022). For this reason, different variations of SMOTE are proposed to address these shortcomings. According to Thejas et al., more than 85 variants of SMOTE have been proposed (Thejas et al. 2022). Most of these SMOTE variants mainly address data complexity problems, including noise, class overlap, and small disjuncts to improve synthetic data quality and classification performance (Garcia et al. 2015; Vuttipittayamongkol et al. 2021; Nekooimehr and Lai-Yuen 2016; Almutairi and Janicki 2020). Some of the recent variants grouped by the complexity they address are discussed in detail below.

SMOTE for noise: An improved SMOTE variant is proposed by Asniar et al (Maulidevi and Surendro 2022) to identify noisy samples in minority-class regions by using

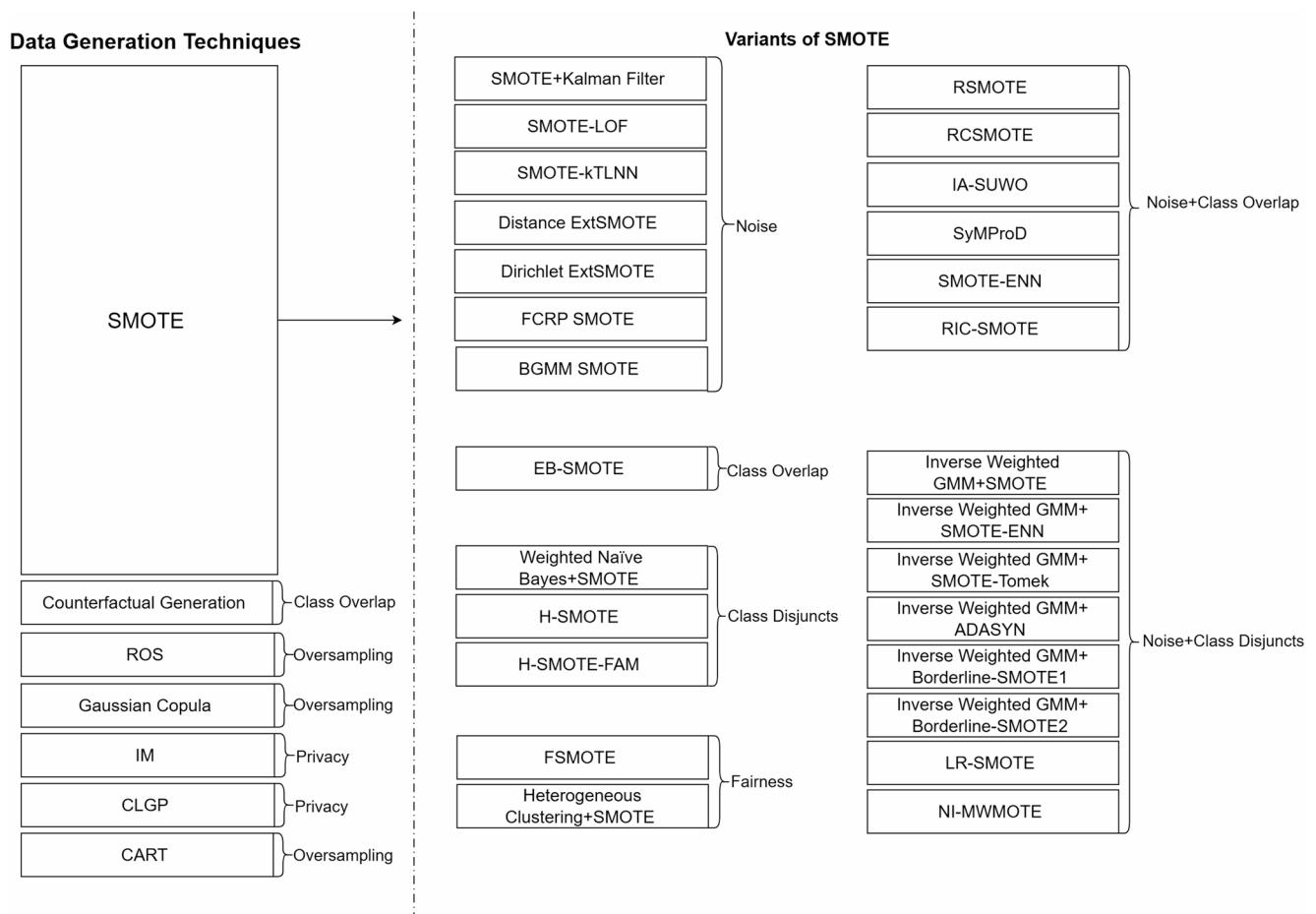


Fig. 7 Overview of data generation techniques and corresponding data complexity addressed

the Local Outlier Factor (LOF). To calculate LOF, the nearest neighbours were determined using K-Nearest Neighbour (KNN). Then, k -distance and reachability distance were calculated to estimate local reachability density and LOF value. The noisy samples were then eliminated and not used in oversampling. The optimal value of k is important to determine a sample as noise, as it directly affects the quality of synthetic samples. Also, G.S. et al. (Thejas et al. 2022) extended the traditional SMOTE by combining it with the Kalman filter to filter noisy samples after balancing the dataset via SMOTE. The technique was only applied to binary datasets. The computational complexity of the Kalman filter is cubic in size, which is an overhead when applied to high-dimensional datasets. Furthermore, SMOTE-kTLNN is a hybrid SMOTE variant, based on traditional SMOTE and a two-layer nearest neighbour classifier (Sun et al. 2024). In this method, the dataset is balanced using the traditional SMOTE algorithm. Afterwards, the balanced dataset is divided into n equal parts. On each part, a two-layer nearest neighbour classifier (kTLNN) is trained. The trained kTLNN is then used to predict the whole data by combining its outputs with n classifiers. Samples most frequently misclassified using the majority voting rule are considered noise and removed from the final dataset. The classification performance has significantly improved with SMOTE-kTLNN, but it has high computational complexity and involves more hyperparameters compared to SMOTE.

In addition, four different variants of SMOTE, including Distance ExtSMOTE, Dirichlet ExtSMOTE, FCRP SMOTE, and BGMM SMOTE, are proposed to handle noisy samples in minority class (Matharaarachchi et al. 2024). All these variants are based on weighted averages of nearest neighbours. To assign weights to nearest neighbours, Distance ExtSMOTE uses inverse distance between median centroid and k neighbours, Dirichlet ExtSMOTE uses a probabilistic framework, FCRP SMOTE uses Chinese Restaurant Process (CRP), and BGMM is based on clustering techniques where each sample is assigned to a specific cluster. Among all of them, Dirichlet ExtSMOTE outperforms most of the other proposed and existing state-of-the-art methods. The experiments are carried out on small and binary datasets.

SMOTE for class overlap: Sun et al. presented a new variant to handle class overlap called Expandable Borderline SMOTE (EB-SMOTE). EB-SMOTE divides the samples into three regions, i.e., noise, borderline, and safe regions, and then generates synthetic samples for both minority and majority classes in their respective borderline regions. The result improves with the generation of borderline instances, but the effect on model bias toward the majority class needs to be studied in detail by assessing other evaluation metric

scores as the samples are generated for both minority and majority classes (Sun et al. 2025).

SMOTE for class disjuncts: Sağlam and Cengiz (2024) proposed a system that combines a weighted Naïve Bayes algorithm with SMOTE. The weighted Naïve Bayes algorithm is used to locate imbalanced local regions within k -nearest neighbours (knn) of each query point. After identifying them, the resampling rate and estimation of the minority data samples in each region are determined. Random oversampling (ROS) and SMOTE were used to generate new samples in each local region. However, while applying SMOTE, the neighbours to be connected were selected randomly which may cause some links to pass through the majority class regions, resulting in noise generation. In addition to that, the performance of the method is highly sensitive to the selection of the k value. A smaller value tends to overfit the model, and a larger one can lead to flexibility loss. In another study, a new SMOTE variant named H-SMOTE-FAM, which combines histogram and a supervised Adaptive Resonance Theory (ART), i.e., fuzzy ARTMAP (FAM) is presented (Liaw et al. 2025). In this technique, the minority class points are assigned to different histogram bins based on position and range intervals. The samples to be generated for each sample bin are calculated using the number of majority samples and available bins. Then, traditional SMOTE algorithms are applied to generate synthetic samples. To handle increased data size, FAM encodes incoming data into a multi-dimensional representation to self-organise data samples compactly. FAM serves as an incremental learning classifier to learn patterns in a compact form from a balanced dataset. The chance of generating redundant minority class samples is reduced using this method. H-SMOTE-FAM outperforms state-of-the-art SMOTE variants and Improved GAN (I-GAN).

SMOTE for noise and class overlap: Chen et al. (2021) presented an improved version of SMOTE called self-adaptive robust SMOTE (RSMOTE) to generate synthetic samples. This method uses relative density to calculate the local density of each minority data point and classify them into borderline and safe region samples. The new samples were then generated in borderline and safe regions only in order to balance the dataset. Another study presented Range-Controlled SMOTE (RCSMOTE) which uses a sample categorisation scheme to locate the minority class samples that are suitable for oversampling. Also, to generate samples only in safe regions and to prevent their penetration in the majority class regions, a range calculation method was used based on the characteristics of the input data. However, it was only evaluated on binary-class problems (Soltanzadeh and Hashemzadeh 2021). Additionally, Wei et al. (2020) proposed an improved adaptive semi-supervised weighted oversampling (IA-SUWO) technique. IA-SUWO tried to maximise

the probability of extracting all important minority class data samples to generate more robust artificial data samples. It was based on least square support numerical spectrum values and the Improving Majority Weighted Minority Oversampling (IMWMO) method to assign weight to minority data points. The performance of IA-SUWO deviates when applied to high-dimensional datasets. Another oversampling technique, namely Synthetic Minority based on Probabilistic Distribution (SyMProD) is proposed by Kunakorntum et al. (2020). This technique is based on a z-score normalisation score that is used for comparison with a noise threshold to remove noise and the probability distribution of the minority class to reduce the class overlap problem. The method needs to define the optimal value of four parameters to perform oversampling, including noise threshold, cutoff threshold, K-nearest neighbour, and minority nearest neighbour, which may vary with each dataset. Moreover, Sasada et al. (2020) combined two well-known SMOTE variants called SMOTE-Edited Nearest Neighbour (SMOTE-ENN) and SMOTE-Tomek to handle both class overlap and noisy samples. Tomek links, defined as pairs of nearest neighbours from different classes with minimum Euclidean distance, tend to remove minority samples from the borderline region, potentially decreasing the number of minority class samples. Also, ENN may remove potential borderline minority samples in the small-sample imbalanced dataset where the minority class has a lower local distribution.

Further, Region-based Information Contribution Oversampling (RIC-SMOTE) is a new oversampling technique that leverages information characteristics of samples in different regions to generate synthetic samples (Xie et al. 2025). The technique uses the K-means algorithm to divide the minority class samples into different regions, and then the number of synthetic samples to be generated is determined by the contribution of each sub-region after removing noisy samples. To find the optimal k value, it employs the Differential Evolution (DE) algorithm. The technique is evaluated using only one classifier, i.e. Random Forest (RF).

SMOTE for noise and class disjuncts: Xing et al. (2022) combined inverse weighted Gaussian Mixture Model (GMM) clustering with SMOTE, SMOTE-ENN, SMOTE-Tomek, Adaptive Synthetic Sampling (ADASYN), Borderline-SMOTE1 and Borderline-SMOTE2 to predict diffuse large B-cell lymphoma (DLBCL). Among all, inverse-weighted GMM performed best with SMOTE-ENN while using a Multi-Kernel Framework (MKL) for training and testing. The SMOTE-ENN is a process that removes the generated samples from the dataset if half of their neighbours do not belong to the same class. The performance of the GMM clustering algorithm reduces when the data has skewness or deviation from Gaussian shapes. The SMOTE-ENN

may not perform well for highly overlapped classes, and it is also computationally expensive for high-dimensional and large datasets because it removes noisy samples after applying SMOTE. Additionally, (Liang et al. (2020) proposed an improved version of SMOTE based on k -means and Support Vector Machine (SVM) named LR-SMOTE. Firstly, an SVM classifier is trained on the dataset to determine the k -nearest misclassified minority class neighbour. The average distance is calculated by using Euclidean distance to find the resampling rate to generate new data samples. The performance of LR-SMOTE depends highly on the k -value and SVM classifier. Furthermore, Wei et al. (2020) presented a Noise-Immunity Majority-Weighted Oversampling Method (NI-MWMOTE). It uses a flexible noise-handling scheme that combines Euclidean distance with neighbour density to find the suspected noise probability identified by knn as true noise. Afterwards, Aggregative Hierarchical Clustering (AHC) was used to cluster minority class samples to prevent ignoring class disjuncts and to determine sampling size and then the weighted Minority Oversampling Technique (MWMOTE) was used to generate synthetic samples. NI-MWMOTE is sensitive to the selection of k -value and noise threshold.

SMOTE for fairness: Chakraborty et al. (2021) proposed a SMOTE variant that considers fairness with oversampling called Fair-SMOTE (FSMOTE). It divides the dataset based on protected attributes. Protected attributes can be defined as groups that split the whole dataset into two groups, i.e., privileged and unprivileged. Then, FSMOTE oversamples by creating new points near the parent data point. After oversampling, biased points are removed from the training dataset using fair situation testing, i.e., flipping the protected attribute value and analysing if there is any difference in the outcome; if yes, that data point is removed from the training dataset. The results are better when compared with other methods, including Fairway and off-the-shelf logistic regression. In another study, Sonoda (2023) mentioned that FSMOTE cannot generate diverse synthetic data because it uses homogeneous clusters for oversampling. To address this issue, Sonoda et al. incorporate sampling from heterogeneous clusters to generate class-mix and group-mix features. The instances from these heterogeneous over-sampled clusters are then used to balance the dataset. This approach is more effective in generating diverse data points that contribute valuable information to the dataset.

Apart from SMOTE and its variants, some of the other data generation techniques are defined below:

3.2.2 Counterfactual generation

Unlike other techniques, Wang et al. (2023) proposed a method that generates new minority class samples using

majority class data points called counterfactuals. To enhance the decision boundary (class overlap), the majority class data points were randomly selected and perturbed using truncated probability distribution unless it is classified as a minority class sample. The method can generate information-rich data points as it uses the majority class distribution. Therefore, this method can be used in cases where minority samples are not enough for further generation. However, the random selection of the majority class data points increases the overall computational cost to perform perturbations for each feature value for high-dimensional datasets.

3.2.3 Random oversampling

ROS is a simple technique to expand a dataset. It increases the data size by randomly sampling from minority classes using replacement (Figueira and Vaz 2022). For example, Batuwita and Palade (2010) used ROS in order to balance the occurrence of each class in the dataset. The results show that the classification results improved with ROS, but this technique only replicates the existing samples rather than generating synthetic samples, therefore, it may lead to overfitting, and it does not add any meaningful information to the dataset. Additionally, ROS only increase data volume and does not consider data distribution properties while oversampling.

3.2.4 Gaussian Copula

Gaussian Copula can be extended to generate synthetic data. It models the data attribute dependence in the dataset. A copula with the corresponding marginal distribution and a multivariate Gaussian joint distribution function is called a Gaussian copula (Pathare et al. 2023). The joint distribution obtained from the Copula function can be used to generate synthetic data samples. A comparison between different tabular synthetic data generation techniques, including the Gaussian copula, is drawn in a study by Pathare et al. (2023). The results show that Gaussian copula cannot capture the tail dependencies among extreme values that affect the quality of the synthetic data (Benali et al. 2021).

3.2.5 Independent marginal

Independent Marginal (IM) is a synthetic data generation technique that samples from the empirical marginal distributions of each variable. The empirical marginal distribution is computed using observed data (Goncalves et al. 2020). The utilisation of IM for synthetic data generation in order to preserve patient privacy shows that IM performs better in identity disclosure metrics as it does not model the

dependence between variables, which can be used to predict unknown attributes (Goncalves et al. 2020). This method has a low computational cost, and the marginal distribution estimation can be done in parallel. As IM does not capture the statistical dependencies between variables, that may fail to reflect the hidden structure of real data.

3.2.6 Categorical latent gaussian process

The Categorical Latent Gaussian Process (CLGP) is used to generate multivariate categorical data. In order to get categorical values, CLGP maps the points in a lower-dimensional continuous latent space to probabilities (by softmax) via non-linear transformations (Dunson and Xing 2009). Richer latent non-linear mappings in CLGP enable the depiction of extremely intricate whole joint distributions. The study (Goncalves et al. 2020) employs CLGP for synthetic medical data generation. The results show that private information leakage using CLGP is not significant and it was capable of capturing relationships between variables to generate realistic data. However, CLGP is non-conjugate; inference is significantly more complicated than other models. It is necessary to use an approximation Bayesian inference technique like Variational Bayes (VB). Therefore, the CLGP inference scales poorly with data size (Goncalves et al. 2020).

3.2.7 Classification and regression trees

Classification and Regression Trees (CART) is an ML method that is extensively used to generate synthetic data. CART is an algorithmic modelling technique that recursively splits a dataset into different groups based on homogeneous outcomes. The splits are defined as yes-or-no questions with reference to the predictor space, and they can be visualised as a tree structure. Each final group's values (leaf) roughly correspond to the conditional distribution of the expected variable for units whose predictors satisfy the requirements for that group. The artificial values are produced through sampling from a suitable data subset (Nowok et al. 2016). Rankin et al. (2020) utilise CART to generate synthetic samples. The numerical attributes were synthesised using normal linear regression, whereas categorical and binary categorical variables were synthesised using polytomous logistic regression and logistic regression, respectively. CART has better performance metric scores when compared with multiple data generation techniques. However, generating trees can be computationally expensive when applied to large datasets.

3.3 RQ3. How have data generation techniques performed?

A comparison has been drawn between different data generation techniques, which are validated on common medical datasets and evaluation metrics. These datasets include the Pima, Haberman, Newthyroid1, Parkinson disease, and WDBC datasets. All these datasets present classification problems. A summary of these datasets is shown in Table 1.

Figure 8 shows the Area Under Curve (AUC) and Geometric Mean (Gmean) comparison on the Pima, Haberman, Newthyroid1, Parkinson disease, and WDBC datasets of different oversampling techniques. The comparison does not include all studies discussed in the paper due to the non-standardised evaluation method.

We can see that SMOTE-kTLNN has the highest AUC and G-mean values for the Pima and Haberman datasets compared to other methods, with a slightly low G-mean score on Newthyroid1 dataset. For the Parkinson dataset, NI-MWMOTE achieves the highest AUC and Gmean scores compared to RCMOTE and IA-SUWO. However, RIC-SMOTE has the highest scores on the WDBC dataset, while the AUC scores of the other two techniques are similar, with a slight increase in Gmean score for the counterfactual technique. Overall, SMOTE-KTLNN demonstrates better performance on all datasets for which it is compared. NI-MWMOTE and RIC-SMOTE perform better for Parkinson and Wisconsin datasets, respectively. All these three methods incorporate a noise removal mechanism before or after oversampling, which shows that handling noisy samples can help improve the classification performance. However, we shouldn't conclude SMOTE-kTLNN to be better as the comparison on different datasets does not include the same techniques. Moreover, the comparative analysis remains limited due to the use of inconsistent evaluation metrics across studies, which results in the exclusion of a substantial number of methods addressing other data complexities.

Consequently, the difference in the performance of these techniques on each dataset depends on several other factors, including the Imbalance Ratio (IR), distribution of minority and majority class samples, and the data complexity handled by each technique. For example, the number of samples generated in the class overlap region has a greater effect on classification performance as compared to data samples generated in the safe minority region. Therefore, techniques

such as RIC-SMOTE and RSMOTE identify and try to enhance class overlap in borderline regions have a better performance as compared to others. Apart from that, the effectiveness of the noise detection method directly influences the quality of the generated samples, as undetected noisy data points can increase the number of outliers in the dataset, which adversely affects the performance of a classifier. For example, methods such as SMOTE-kTLNN and LR-SMOTE achieved high performance scores when compared with others on the Haberman dataset. Notably, both these methods incorporate a noise detection and removal mechanism after balancing the dataset. Moreover, there is no standardised set of criteria for evaluating synthetic data. Different studies used varying evaluation metrics, making direct comparison challenging and preventing any definitive conclusion about one method outperforming the others. Lastly, the difference in the underlying experimental setup of each technique may also result in different performance scores.

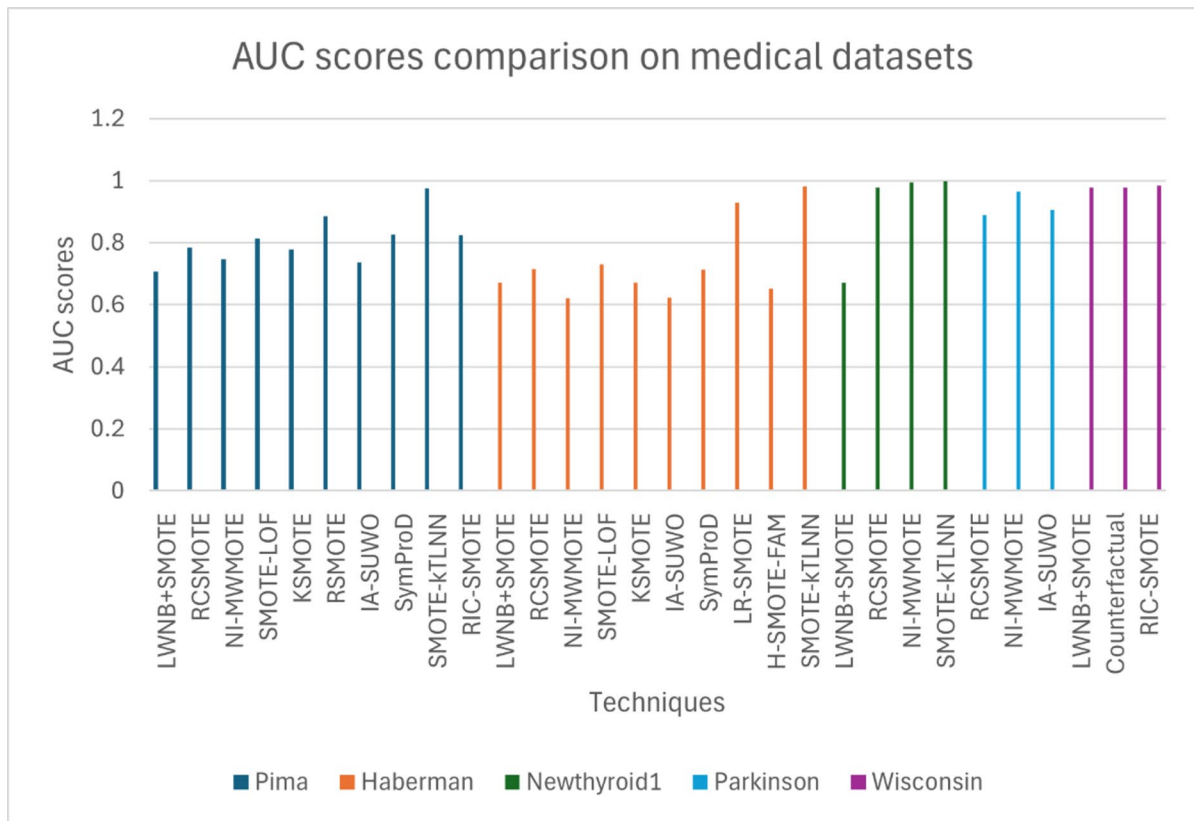
In conclusion, the above comparisons show that synthetic data generation methods help improve the performance of the AI models by improving the data quality. However, each method has its own strengths and limitations. There is no single method that outperforms others in all evaluation metrics.

4 Discussion and findings

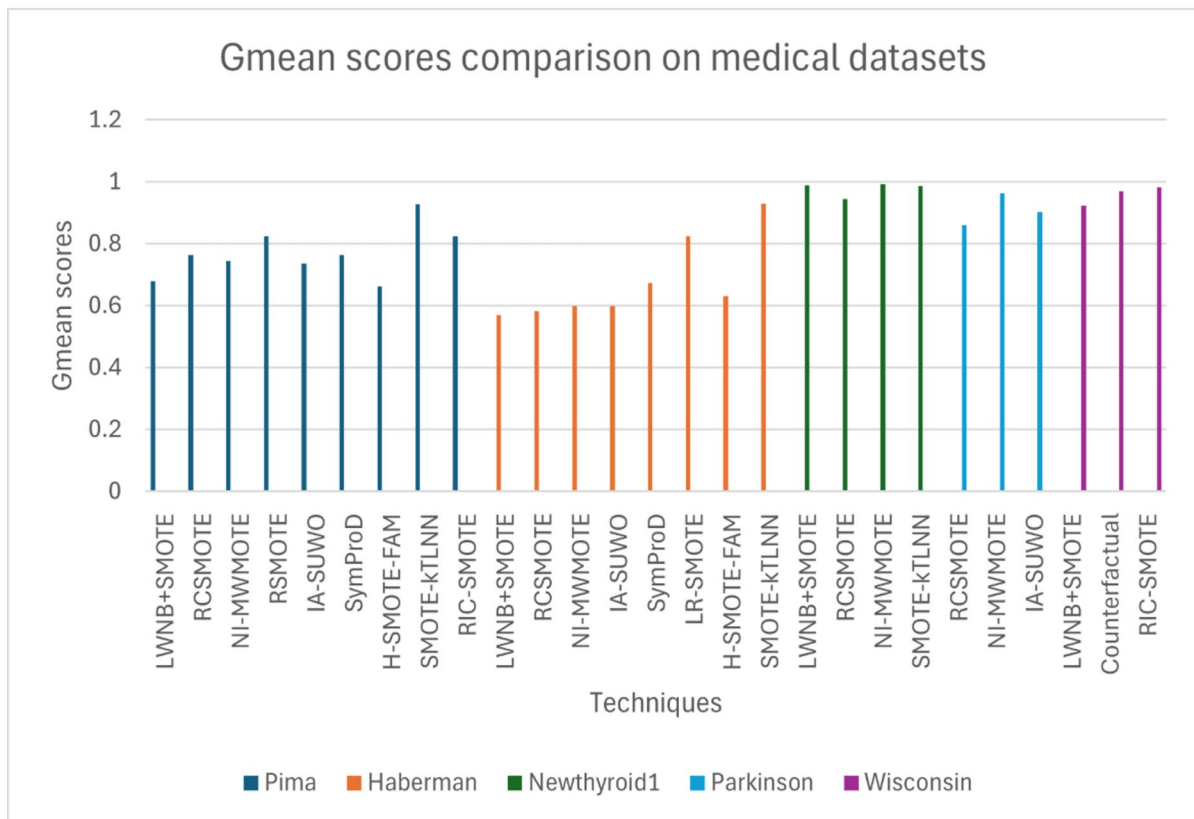
Due to the inaccessibility of medical data and corresponding concerns about data quality, synthetic data generation is of great interest to realise the full potential of data-driven AI technologies (Figueira and Vaz 2022). Synthetic data generation can help improve the quality of medical data by augmenting the dataset with newly generated artificial samples that exhibit statistical properties of real-world data in order to balance the representation of all groups or classes. Synthetic data can also help in ensuring consistency across data formats and labels. Different techniques are proposed in the literature to generate synthetic medical data. These techniques handle different data complexities, such as identifying and removing outliers, reducing class overlap to enhance class boundaries, and addressing disjuncts to better represent rare conditions, thereby improving the accuracy and completeness of datasets. A total of 21 studies

Table 1 Summary of the most commonly used medical datasets

Dataset	Number of classes	Number of samples	Dimension	Majority class	Minority class	Imbalance ratio (IR)
Pima	2	768	8	500	268	1.866
Haberman	2	306	3	225	81	2.778
Newthyroid1	3	215	5	180	35	5.143
Parkinson disease	2	195	23	147	48	3.060
WDBC	2	569	30	357	212	1.7



(a)



(b)

Fig. 8 AUC and Gmean comparison across different medical datasets **a** AUC scores comparison **b** Gmean scores comparison

employed SMOTE or its variants in conjunction with other methods, including GMM, Naïve Bayes, SVM, and Kalman filter to generate artificial data. 2 studies provide an analysis of different synthetic data generation tools, including Synthea based on PADARSER framework, Synthpop utilises CART, Simpop employs Logistic Regression (LR) and Datasynthesizer uses Bayesian Network. The remaining 4 studies discuss Gaussian Copula, CART, CLGP, IM and counterfactual generation. Table 2 represents the overview of commonly used data generation techniques.

Among all of them, SMOTE is a widely accepted and commonly used standard technique (Azhar et al. 2022). A significant amount of work is still ongoing to further improve SMOTE, as highlighted by our literature review. There are multiple variants proposed in the literature to overcome the limitations of traditional SMOTE. A study conducted by Azhar et al. outlining the analysis of different SMOTE variants for similarities, differences and data complexities supported with a series of experiments performed on 24 binary datasets concludes that SMOTE variants have improved the classification performance by up to 12% and can reduce data complexity (Azhar et al. 2022). The study also highlights that SMOTE variants are useful in practical applications, and the development of better SMOTE-based algorithms can enhance classification performance on datasets that are imbalanced and complex.

Moreover, methods like FSMOTE and Oversampling using heterogeneous clusters can be extended to medical datasets to mitigate imbalances in protected attributes that may otherwise lead to bias thus enhancing data completeness (Chakraborty et al. 2021; Sonoda 2023). Heterogeneous clustering and counterfactual generation are some of the novel techniques that not only balance the dataset but also ensure the generation of diverse data points that add useful information to the dataset (Wang et al. 2023; Sonoda 2023).

This highlights the gap that a large part of standard synthetic data generation models is not designed for medical datasets. Studies by Azizi et al. (2021) and Goncalves et al. (2020), which are focused on medical datasets, only consider privacy preservation as the main goal. The results show a tradeoff between performance and sensitive information disclosure metrics. Therefore, several de-anonymisation strategies at pre-processing stages can help in privacy preservation with minimal information loss and better data utility (Onesimu et al. 2022). While most SMOTE variants primarily address class imbalance by handling data complexities, methods such as FSMOTE and fair class balancing prioritise enhancing fairness through synthetic data generation but tend to overlook data complexities. Furthermore,

the comparison of different techniques in Fig. 8 on common medical datasets shows that the performance of each data generation technique depends on the characteristics and distribution of a dataset, IR, noise detection method and data complexity addressed by each technique. Therefore, there is no agreement on any one method being better than others. Additionally, as shown in Table 3, the studies reviewed have employed varying sets of evaluation metrics. While metrics such as G-mean and AUC are commonly used to assess classifier performance on synthetically balanced datasets, they cannot be considered standard measures, as their use is not consistent across all studies. This inconsistency makes it difficult to compare different oversampling methods. Therefore, standardising the evaluation process of oversampling techniques is essential to enhance their practical applicability, particularly in medical AI. The standardised evaluation framework may include metrics that assess the utility, fidelity, privacy and risk, fairness and bias scores of synthetic data.

Thus, our literature review highlights the need to investigate oversampling techniques for synthetic medical data generation. Bridging this gap requires addressing both class and protected attribute imbalances, as well as accounting for data complexities while developing strategies to generate diverse synthetic data points. Achieving this goal requires validation on diverse medical datasets across a wide range of evaluation metrics. These evaluation metrics should be a blend of model performance indicators, statistical testing, and fairness assessment metrics. Additionally, AI models developed on synthetic data can be tested in a real-time clinical setting for user evaluation to identify potential challenges beyond accuracy, precision, recall, etc, to uplift the trustworthiness of the medical applications or software.

5 Conclusion and future work

This article reviews the synthetic data generation process and recent research studies to investigate different data generation techniques for medical datasets. For this purpose, we explored five different databases including PubMed, ACM, IEEE, ScienceDirect and Google Scholar. A total of 27 articles were selected and further analysed for our proposed research questions based on inclusion and exclusion criteria. Findings from this study can help the development of effective AI systems in healthcare settings. Hence, our findings lead us to the following insights:

- The reviewed techniques have been tested and evaluated on various datasets, but none of them consistently outperforms the others across all datasets on which they are validated. This highlights that the effectiveness of data

Table 2 Overview of commonly used data generation techniques

Techniques	Strength	Weaknesses
ROS	It increases the dataset size and balances the number of samples for each class	It only replicates the minority class samples, which may lead to overfitting and make it less feasible for practical use
SMOTE	It is a widely used synthetic data generation technique. It tends to generate new samples by interpolating between existing samples and their neighbours	The traditional SMOTE algorithm overlooks the underlying distribution and complexities of datasets, which can hinder its ability to generalise effectively across diverse datasets and may consequently degrade the performance of classifiers trained on such datasets. Ongoing research aims to enhance the performance of traditional SMOTE by introducing different variants. Some of these variants also incorporate fairness by balancing both sensitive attributes and class labels, making them suitable for medical datasets
CART	CART is a simple technique that splits a dataset into different groups that can be visualised as a tree structure. It can handle any type of data. Synthpop (Goncalves et al. 2020; Rankin et al. 2020) is a tool that uses CART model to generate synthetic data	CART generate trees that increase complexity and computational cost for real-world large datasets
Counterfactual generation	Counterfactual generation technique uses majority class samples to generate minority ones, incorporating data distribution into the generation process. It is suitable when minority class samples are insufficient or lack enough information for subsequent modelling	The classification of a generated sample as belonging to the minority class requires substantial empirical validation for practical use. Also, the selection of the majority class sample for counterfactual generation is a challenging task

generation methods is highly dependent on factors such as the IR, data distribution, and complexity.

- There are different data generation methods proposed that leverage heterogeneous clustering and majority class distribution information that not only increase the dataset size but also tend to generate information-rich samples (Wang et al. 2023; Sonoda 2023).
- Most of the techniques consider data complexities while designing the data generation process, and some of them address imbalance in protected attributes to mitigate the risk of bias, which highlights the need to investigate oversampling techniques for the generation of synthetic medical data that implements a hybrid approach by considering both the class and protected attribute imbalances along with data complexities.
- Each study employs different evaluation metrics, underscoring the need for comprehensive and standardised measures to assess the quality of generated data samples and to compare different data generation techniques. Considering the current paradigm shift and the increasing use of AI for synthetic data generation, establishing regulations to standardise the evaluation of synthetic data quality in medical AI could provide a foundational framework for regulating data augmentation practices.
- The standardisation may include an evaluation framework of synthetic data consisting of metrics that assess different aspects such as utility, fidelity, privacy and risk, fairness and bias.

- In addition to controlled evaluation methods, AI models trained on synthetic data can be evaluated in real clinical settings to assess their performance on real-world medical datasets. Such evaluations can offer valuable insights into the effectiveness of synthetic data and help identify areas for future improvement.
- Real-world clinical evaluation will also facilitate feedback from key stakeholders such as doctors, clinicians, and patients. Such collaborations will help refine the AI technology to better align with user needs, particularly in terms of accuracy and explainability, which are essential for building trust.

In addition to the limitations of our study, our inclusion and exclusion criteria may have introduced bias, as we only considered studies evaluated on at least one medical dataset. Also, we have not included some other databases, such as Scopus and Web of Science, which limits the scope of the study. In future, we will expand our search by formulating additional research questions to broaden the scope of selected articles and incorporating other databases, especially Scopus and Web of Science, to further enhance our findings. Also, other data modalities such as images and free-text are widely used in the healthcare industry for disease diagnosis. Therefore, in future, we will extend our review to data generation techniques for other data modalities to broaden the scope of this study within the context of medical AI. In addition, we plan to investigate studies on

synthetic data evaluation frameworks that can help shape the standardisation process.

Appendix A

Tables 3, 4.

Table 3 Summary of recent data generation techniques where CD=Class Disjuncts, CO=Class Overlap, N=Noise, BO=Balance class with Oversampling, F=Fairness, and P=Privacy

Ref	Data complexity	Generation technique	Evaluation method
Sağlam and Cengiz (2024)	CD	Random Oversampling (ROS) and SMOTE combined with locally weighted Naïve Bayes (LWNB)	Geometric Mean (G-mean), F1-score, Area Under Curve (AUC), Mathews Correlation Coefficient (MCC)
Nithya et al. (2024)	CO	Support Vector Machine (SVM)-SMOTE + Kernel Density Estimation (KDE)	Accuracy, Precision, Recall, F1-score, AUC
Pathare et al. (2023)	BO	Classification and Regression Trees (CART), Generative Adversarial Networks (GANs), Gaussian Copula, Conditional Tabular GAN (CTGAN), Triplet-based Variational Autoencoder (TVAE), Bayesian Network, Recurrent Neural Network (RNN), Copula GAN	Propensity score, Cluster log metric, Prediction accuracy, Execution time
Wang et al. (2023)	CO	Counter-factual sampling using perturbation	AUC, G-mean
Rodriguez-Almeida et al. (2022)	CO, CD	SMOTE, Borderline SMOTE, ADASYN	Maximum Mean Discrepancy (MMD), KL Divergence
Sonoda (2023)	F	Oversampling based on heterogeneous clusters	Balanced accuracy, SP, EOD, Equal Opportunity Difference (EOD), Statistical Parity Difference (SPD), Equalised Odds (E. Odds)
Maldonado et al. (2022)	BO	Feature-weighted SMOTE (FW-SMOTE) based on Induced Minkowski Ordered Weighted Averaging Distance (IMOWAD) operator	Ranking, AUC, Holm test, G-mean
Maulidevi and Surendro (2022)	N	SMOTE based on Local Outlier Factor (SMOTE-LOF)	Accuracy, Precision, Recall, F-measure, AUC
Thejas et al. (2022)	N	Kalman filter combined with SMOTE (KSMOTE)	AUC, Accuracy, Recall, F1-score, Precision
Xing et al. (2022)	N, CD	Inverse weighted Gaussian Mixture Model (GMM)+ SMOTE-ENN+Multi Kernel Learning (MKL)	AUC, Accuracy, Recall, Precision, F1-score, Expected Calibration Error (ECE), Brier
Azizi et al. (2021)	P	Sequential decision trees (conditional trees)	Information-theoretic, Kaplan-Meier curve, Logrank test, Goodman-Kruskal tau statistic
Benali et al. (2021)	BO	Gaussian Copula, T-Copula	Akaike Information Criterion (AIC), Root Mean Square Error (RMSE), Kolmogorov Smirnov test, Kendal method
Chen et al. (2021)	N, CO	Self-adaptive Robust SMOTE (RSMOTE)	Friedman test, Holm's test, Precision, Recall, AUC, F1-score
Soltanzadeh and Hashemzadeh (2021)	N, CO	Combines sample categorisation scheme and RC-SMOTE	AUC, Accuracy, G-mean
Chakraborty et al. (2021)	F	Fair-SMOTE which groups the dataset based on protected attributes and class	Recall, Precision, Accuracy, F1-score, Average Odds Difference (AOD), EOD, SPD, Disparate Impact (DI), Scott-Knott test
Goncalves et al. (2020)	P	Synthpop, SimPop, Synthea, Independent Marginal (IM), Bayesian Network, Mixture of Product of Multinomials (MPoM), Categorical Latent Gaussian Process (CLGP), GAN, Multiple Imputation	KL divergence, correlation, difference, log cluster support coverage metric, membership disclosure, SEER edits
Rankin et al. (2020)	P	Synthpop, Datasynthesizer, SimPop, Synthea, CART, Bayesian	Minimum leaf size, smoothing, normalised mutual information
Liang et al. (2020)	N, CD	Limiting the Radius (LR-SMOTE)	G-mean, ROC curve, Accuracy, Recall
Wei et al. (2020)	N, CO	Combines adaptive semi-supervised clustering with Majority Weighted Minority Oversampling Technique (MWMOTE) called IA-SUWO	Sensitivity, Specificity, G-mean, AUC
Kunakorntum et al. (2020)	N, CO	Data generation based on probability distribution (SyMProD)	AUC, G-mean, F-measure
Sasada et al. (2020)	N, CO	Combines SMOTE with Edited Nearest Neighbour (SMOTE-ENN) and SMOTE-Tomek	Accuracy, F-measure

Table 3 (continued)

Ref	Data complexity	Generation technique	Evaluation method
Wei et al. (2020)	N, CD	MWMOTE with adaptive noise and sub-cluster (NI-MWMOTE)	AUC, Sensitivity, Accuracy, G-mean, F-measure
Liaw et al. (2025)	CD	Histogram SMOTE-based oversampling combined with fuzzy ARTMAP (FAM) network (H-SMOTE-FAM)	G-mean, AUC, F-measure, Wilcoxon signed-rank test
Sun et al. (2024)	N	SMOTE combined with two-layer nearest neighbour classifier (SMOTE-kTLNN)	Precision, Recall, F1-measure, G-mean, Wilcoxon test
Matharaarachchi et al. (2024)	N	Distance ExtSMOTE, Dirichlet ExtSMOTE, FCRP SMOTE, and BGMM SMOTE	Precision, Recall, F1-measure, PR-AUC, MCC, Friedman test, Post hoc pairwise test
Xie et al. (2025)	N, CO	Region-based Information Contribution Oversampling (RIC-SMOTE)	True Positive Rate (TPR), True Negative Rate (TNR), G-mean, AUC, one-tailed Wilcoxon signed rank test
Sun et al. (2025)	CO	Expandable Borderline SMOTE (EB-SMOTE)	Accuracy, Precision, PRAUC, AUROC, Wilcoxon-signed rank test

Table 4 An overview of medical datasets used in each selected article

References	Medical datasets
Sağlam and Cengiz (2024)	Newthyroid1, Wisconsin Breast Cancer Dataset (WDBC), Pima Indian Diabetes Dataset (Pima), Kyphosis, SAheart, ecoli1, Haberman
Nithya et al. (2024)	Cerebral dataset stroke dataset
Pathare et al. (2023)	Lung cancer, Breast cancer, Lymphography, Statlog (heart)
Wang et al. (2023)	WDBC, inhouse lung cancer dataset
Rodriguez-Almeida et al. (2022)	Alzheimer (MNCD), Alzheimer* (MNCDRED), Diabetes Bangladesh (BANG), Early Diabetes (EarlyDM), Heart Disease (HeartDis), Kidney Chronic Disease, Pima, South Africa Cardio
Sonoda (2023)	Heart dataset
Maldonado et al. (2022)	Ecoli variants, Glioma, Lung, Srbct dataset
Maulidevi and Surendro (2022)	Pima, Haberman
Thejas et al. (2022)	Haberman, Hepatitis, Contraceptive Method Choice (CMC), Pima, Z-Alizade sani dataset
Xing et al. (2022)	Shanxi cancer hospital dataset
Azizi et al. (2021)	Oncology trial N0147 dataset
Benali et al. (2021)	Breast cancer, WDBC
Chen et al. (2021)	Breast tissue2, Ecoli, Heart, Pima, Breast cancer, Skin dataset
Soltanzadeh and Hashemzadeh (2021)	Bupa, Pima, Breast, Haberman, Newthyroid, Hepatitis, Ecoli, Breast tissue, heart, liver, Parkinsons, transfusion, ILPD
Chakraborty et al. (2021)	Heart Health (protected attribute: Age)
Goncalves et al. (2020)	Cancer registry data (breast, respiratory, non-solid cancer)
Rankin et al. (2020)	WDBC, Chronic kidney disease, Cardiotocography, Dermatology, Diabetic retinography, Heart disease, Lymphography, Stroke, Thoracic surgery, Thyroid disease
Liang et al. (2020)	Haberman, Diabetes, Blood dataset
Wei et al. (2020)	Pima, Parkinson, WDBC, heart, Haberman
Kunakorntum et al. (2020)	Ecoli, Haberman, Heart, Pima
Sasada et al. (2020)	Breast cancer, WDBC, Haberman
Wei et al. (2020)	WDBC, Pima, Parkinsons, Liver, Heart, Haberman, Breast
Liaw et al. (2025)	Haberman, Pima, DiabetesV, Indian Liver Patient Dataset (ILPD), Contraceptive method choice
Sun et al. (2024)	haberman, Pima, new-thyroid1, dermatology6
Matharaarachchi et al. (2024)	thyroid_sick, parkinsons, haberman, breast_cancer
Xie et al. (2025)	newthyroid2, Pima, WDBC
Sun et al. (2025)	sick_euthyroid, thyroid_sick, arrhythmia, mammography

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Declarations

Conflict of interest The authors declare that they have no Conflict of interest.

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